AK Library Management Module - Odoo Documentation

This document explains the complete development of the custom Odoo module "ak_library_management" which manages Books and Categories in a library system.

1. Module Overview

Module Name: ak_library_management

Dependencies: base, web

Purpose: Manage library books and book categories with proper menu, views, and CRUD operations.

2. Module Structure

```
ak_library_management/
├── __init__.py
├── __manifest__.py
├── models/
│   ├── __init__.py
│   ├── book.py
│   └── category.py
├── security/
│   └── ir.model.access.csv
└── views/
    ├── book_views.xml
    ├── category_views.xml
    └── menu.xml
```

3. __manifest__.py

```
{
   'name': 'AK Library Management',
   'version': '1.0',
   'summary': 'Manage library books and categories',
   'category': 'Education',
   'depends': ['base', 'web'],
   'data': [
      'security/ir.model.access.csv',
      'views/book_views.xml',
      'views/category_views.xml',
      'views/menu.xml',
   ],
   'installable': True,
   'application': True,
}
```

4. Models

library.book Model

```
from odoo import models, fields

class LibraryBook(models.Model):
   _name = 'library.book'
   _description = 'Library Book'

   name = fields.Char(string='Book Title', required=True)
   isbn = fields.Char(string='ISBN Number')
   publication_date = fields.Date(string='Publication Date')
   price = fields.Float(string='Book Price')
   pages = fields.Integer(string='Number of Pages')
   description = fields.Html(string='Book Summary')
   image_1920 = fields.Image(string='Book Image')
   category_id = fields.Many2one('book.category', string='Category')
```

book.category Model

```
from odoo import models, fields

class BookCategory(models.Model):
    _name = 'book.category'
    _description = 'Book Category'

    category_name = fields.Char(string='Category Name', required=True)
```

5. Security Access Rights

```
id,name,model_id:id,group_id:id,perm_read,perm_write,perm_create,perm_unlink
access_library_book,library.book,model_library_book,,1,1,1,1
access_book_category,book.category,model_book_category,,1,1,1,1
```

6. Views

views/book_views.xml

```
<odoo>


    <!-- TREE VIEW -->

    <record id="view_library_book_tree" model="ir.ui.view">

        <field name="name">library.book.tree</field>

        <field name="model">library.book</field>

        <field name="arch" type="xml">

            <tree>

                <field name="name"/>

                <field name="isbn"/>

                <field name="price"/>

                <field name="pages"/>

            </tree>

        </field>

    </record>
```

```
<!-- FORM VIEW -->
<record id="view_library_book_form" model="ir.ui.view">
  <field name="name">library.book.form</field>
  <field name="model">library.book</field>
  <field name="arch" type="xml">
    <form string="Book">
      <sheet>
        <group>
          <group>
            <field name="name"/>
            <field name="isbn"/>
            <field name="publication_date"/>
            <field name="category_id"/>
          </group>
          <group>
            <field name="price"/>
            <field name="pages"/>
            <field name="image_1920" widget="image"/>
          </group>
        </group>

        <notebook>
          <page string="Description">
            <field name="description"/>
          </page>
```

```
            </notebook>
          </sheet>
        </form>
      </field>
    </record>

    <!-- SEARCH VIEW -->
    <record id="view_library_book_search" model="ir.ui.view">
      <field name="name">library.book.search</field>
      <field name="model">library.book</field>
      <field name="arch" type="xml">
        <search>
          <field name="name" string="Title"/>
          <field name="isbn"/>
        </search>
      </field>
    </record>

    <!-- ACTION -->
    <record id="action_library_books" model="ir.actions.act_window">
      <field name="name">Books</field>
      <field name="res_model">library.book</field>
      <field name="view_mode">tree,form</field>
    </record>

  </odoo>
```

views/category_views.xml

```
<odoo>


  <!-- TREE VIEW -->
  <record id="view_book_category_tree" model="ir.ui.view">
    <field name="name">book.category.tree</field>
    <field name="model">book.category</field>
    <field name="arch" type="xml">
      <tree>
        <field name="category_name"/>
      </tree>
    </field>
  </record>


  <!-- FORM VIEW -->
  <record id="view_book_category_form" model="ir.ui.view">
    <field name="name">book.category.form</field>
    <field name="model">book.category</field>
    <field name="arch" type="xml">
      <form string="Category">
        <sheet>
          <group>
            <field name="category_name"/>
          </group>
        </sheet>
      </form>
    </field>
```

```
    </record>


    <!-- ACTION -->
    <record id="action_book_category" model="ir.actions.act_window">
        <field name="name">Categories</field>
        <field name="res_model">book.category</field>
        <field name="view_mode">tree,form</field>
    </record>


</odoo>
```

view/menu.xml

```
<odoo>

  <!-- Main Menu -->
  <menuitem id="menu_library_root"
        name="Library Management"
        sequence="10"/>

  <!-- Books Menu -->
  <menuitem id="menu_library_books"
        name="Books"
        parent="menu_library_root"
        action="action_library_books"
        sequence="1"/>

  <!-- Category Menu -->
  <menuitem id="menu_library_category"
        name="Category"
        parent="menu_library_root"
        action="action_book_category"
        sequence="2"/>

</odoo>
```

7. Menus & Actions

Main Menu: Library Management

Submenu: Books (Tree & Form view action)

Submenu: Category (Tree & Form view action)

8. Training Objectives Achieved

- Understanding Odoo module structure
- Creating custom models and fields
- Creating Tree, Form, and Search views
- Adding menus and window actions
- Performing CRUD operations